MSG No. 142.564 $I4'_1/ac'd$ [Type III, tetragonal]

Table 1: Wyckoff site: `8a`, site symmetry: `-4'..`

No.	position	mapping
1	$\left[0, \frac{1}{4}, \frac{3}{8}\right]$	`[1,2,13,14]`
2	$\left[0, \frac{1}{4}, \frac{7}{8}\right]$	`[3,4,15,16]`
3	$\left[0, \frac{3}{4}, \frac{5}{8}\right]$	`[5,6,9,10]`
4	$\left[\frac{1}{2}, \frac{1}{4}, \frac{5}{8}\right]$	`[7,8,11,12]`
5	$\left[\frac{1}{2}, \frac{3}{4}, \frac{7}{8}\right]$	`[17,18,29,30]`
6	$\left[\frac{1}{2}, \frac{3}{4}, \frac{3}{8}\right]$	`[19,20,31,32]`
7	$\left[\frac{1}{2}, \frac{1}{4}, \frac{1}{8}\right]$	`[21,22,25,26]`
8	$\left[0, \frac{3}{4}, \frac{1}{8}\right]$	`[23,24,27,28]`

Table 2: Wyckoff site: `8b`, site symmetry: `2.22`

No.	position	mapping
1	$\left[0, \frac{1}{4}, \frac{1}{8}\right]$	`[1,2,3,4]`
2	$\left[0, \frac{3}{4}, \frac{7}{8}\right]$	`[5,6,23,24]`
3	$\left[\frac{1}{2}, \frac{1}{4}, \frac{3}{8}\right]$	`[7,8,21,22]`
4	$\left[0, \frac{3}{4}, \frac{3}{8}\right]$	`[9,10,27,28]`
5	$\left[\frac{1}{2}, \frac{1}{4}, \frac{7}{8}\right]$	`[11,12,25,26]`
6	$\left[0, \frac{1}{4}, \frac{5}{8}\right]$	`[13,14,15,16]`
7	$\left[\frac{1}{2}, \frac{3}{4}, \frac{5}{8}\right]$	`[17,18,19,20]`
8	$\left[\frac{1}{2}, \frac{3}{4}, \frac{1}{8}\right]$	`[29,30,31,32]`

Table 3: Wyckoff site: `16c`, site symmetry: `-1`

No.	position	mapping
1	$[0, 0, 0]$	`[1,5]`
2	$\left[0, \frac{1}{2}, 0\right]$	`[2,6]`
3	$\left[\frac{3}{4}, \frac{1}{4}, \frac{1}{4}\right]$	`[3,7]`
4	$\left[\frac{1}{4}, \frac{1}{4}, \frac{1}{4}\right]$	`[4,8]`
5	$\left[\frac{1}{4}, \frac{3}{4}, \frac{1}{4}\right]$	`[9,29]`
6	$\left[\frac{3}{4}, \frac{3}{4}, \frac{1}{4}\right]$	`[10,30]`
7	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	`[11,31]`
8	$\left[\frac{1}{2}, 0, 0\right]$	`[12,32]`
9	$\left[\frac{3}{4}, \frac{1}{4}, \frac{3}{4}\right]$	`[13,25]`
10	$\left[\frac{1}{4}, \frac{1}{4}, \frac{3}{4}\right]$	`[14,26]`
11	$\left[0, 0, \frac{1}{2}\right]$	`[15,27]`
12	$\left[0, \frac{1}{2}, \frac{1}{2}\right]$	`[16,28]`
13	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	`[17,21]`
14	$\left[\frac{1}{2}, 0, \frac{1}{2}\right]$	`[18,22]`
15	$\left[\frac{1}{4}, \frac{3}{4}, \frac{3}{4}\right]$	`[19,23]`
16	$\left[\frac{3}{4}, \frac{3}{4}, \frac{3}{4}\right]$	`[20,24]`

Table 4: Wyckoff site: `16d`, site symmetry: `2..`

No.	position	mapping
1	$\left[0,\ \frac{1}{4},\ z\right]$	`[1,2]`
2	$\left[0,\ \frac{1}{4},\ \frac{1}{4}-z\right]$	`[3,4]`
3	$\left[0,\ \frac{3}{4},\ -z\right]$	`[5,6]`
4	$\left[\frac{1}{2},\ \frac{1}{4},\ z+\frac{1}{4}\right]$	`[7,8]`
5	$\left[0,\ \frac{3}{4},\ z+\frac{1}{4}\right]$	`[9,10]`
6	$\left[\frac{1}{2},\ \frac{1}{4},\ -z\right]$	`[11,12]`
7	$\left[0,\ \frac{1}{4},\ \frac{3}{4}-z\right]$	`[13,14]`
8	$\left[0,\ \frac{1}{4},\ z+\frac{1}{2}\right]$	`[15,16]`
9	$\left[\frac{1}{2},\ \frac{3}{4},\ z+\frac{1}{2}\right]$	`[17,18]`
10	$\left[\frac{1}{2},\ \frac{3}{4},\ \frac{3}{4}-z\right]$	`[19,20]`
11	$\left[\frac{1}{2},\ \frac{1}{4},\ \frac{1}{2}-z\right]$	`[21,22]`
12	$\left[0,\ \frac{3}{4},\ z+\frac{3}{4}\right]$	`[23,24]`
13	$\left[\frac{1}{2},\ \frac{1}{4},\ z+\frac{3}{4}\right]$	`[25,26]`
14	$\left[0,\ \frac{3}{4},\ \frac{1}{2}-z\right]$	`[27,28]`
15	$\left[\frac{1}{2},\ \frac{3}{4},\ \frac{1}{4}-z\right]$	`[29,30]`
16	$\left[\frac{1}{2},\ \frac{3}{4},\ z\right]$	`[31,32]`

Table 5: Wyckoff site: `16e`, site symmetry: `.2'.`

No.	position	mapping
1	$\left[x,\ 0,\ \frac{1}{4}\right]$	`[1,27]`
2	$\left[-x,\ \frac{1}{2},\ \frac{1}{4}\right]$	`[2,28]`
3	$\left[\frac{3}{4},\ x+\frac{1}{4},\ 0\right]$	`[3,25]`
4	$\left[\frac{1}{4},\ \frac{1}{4}-x,\ 0\right]$	`[4,26]`
5	$\left[-x,\ 0,\ \frac{3}{4}\right]$	`[5,15]`
6	$\left[x,\ \frac{1}{2},\ \frac{3}{4}\right]$	`[6,16]`
7	$\left[\frac{3}{4},\ \frac{1}{4}-x,\ \frac{1}{2}\right]$	`[7,13]`
8	$\left[\frac{1}{4},\ x+\frac{1}{4},\ \frac{1}{2}\right]$	`[8,14]`
9	$\left[\frac{1}{4},\ x+\frac{3}{4},\ \frac{1}{2}\right]$	`[9,19]`
10	$\left[\frac{3}{4},\ \frac{3}{4}-x,\ \frac{1}{2}\right]$	`[10,20]`
11	$\left[x+\frac{1}{2},\ \frac{1}{2},\ \frac{3}{4}\right]$	`[11,17]`
12	$\left[\frac{1}{2}-x,\ 0,\ \frac{3}{4}\right]$	`[12,18]`
13	$\left[\frac{1}{2}-x,\ \frac{1}{2},\ \frac{1}{4}\right]$	`[21,31]`
14	$\left[x+\frac{1}{2},\ 0,\ \frac{1}{4}\right]$	`[22,32]`
15	$\left[\frac{1}{4},\ \frac{3}{4}-x,\ 0\right]$	`[23,29]`
16	$\left[\frac{3}{4},\ x+\frac{3}{4},\ 0\right]$	`[24,30]`

Table 6: Wyckoff site: `16f`, site symmetry: `..2`

No.	position	mapping
1	$\left[x,\ x+\frac{1}{4},\ \frac{1}{8}\right]$	`[1,3]`
2	$\left[-x,\ \frac{1}{4}-x,\ \frac{1}{8}\right]$	`[2,4]`
3	$\left[-x,\ \frac{3}{4}-x,\ \frac{7}{8}\right]$	`[5,23]`
4	$\left[x,\ x+\frac{3}{4},\ \frac{7}{8}\right]$	`[6,24]`

continued ...

Table 6

No.	position	mapping
5	$\left[\frac{1}{2}-x,\ \frac{1}{4}-x,\ \frac{3}{8}\right]$	[7,21]
6	$\left[x+\frac{1}{2},\ x+\frac{1}{4},\ \frac{3}{8}\right]$	[8,22]
7	$\left[-x,\ x+\frac{3}{4},\ \frac{3}{8}\right]$	[9,28]
8	$\left[x,\ \frac{3}{4}-x,\ \frac{3}{8}\right]$	[10,27]
9	$\left[x+\frac{1}{2},\ \frac{1}{4}-x,\ \frac{7}{8}\right]$	[11,26]
10	$\left[\frac{1}{2}-x,\ x+\frac{1}{4},\ \frac{7}{8}\right]$	[12,25]
11	$\left[x,\ \frac{1}{4}-x,\ \frac{5}{8}\right]$	[13,16]
12	$\left[-x,\ x+\frac{1}{4},\ \frac{5}{8}\right]$	[14,15]
13	$\left[x+\frac{1}{2},\ x+\frac{3}{4},\ \frac{5}{8}\right]$	[17,19]
14	$\left[\frac{1}{2}-x,\ \frac{3}{4}-x,\ \frac{5}{8}\right]$	[18,20]
15	$\left[x+\frac{1}{2},\ \frac{3}{4}-x,\ \frac{1}{8}\right]$	[29,32]
16	$\left[\frac{1}{2}-x,\ x+\frac{3}{4},\ \frac{1}{8}\right]$	[30,31]

Table 7: Wyckoff site: 32g, site symmetry: 1

No.	position	mapping
1	$[x,\ y,\ z]$	[1]
2	$\left[-x,\ \frac{1}{2}-y,\ z\right]$	[2]
3	$\left[y+\frac{3}{4},\ x+\frac{1}{4},\ \frac{1}{4}-z\right]$	[3]
4	$\left[\frac{1}{4}-y,\ \frac{1}{4}-x,\ \frac{1}{4}-z\right]$	[4]
5	$[-x,\ -y,\ -z]$	[5]
6	$\left[x,\ y+\frac{1}{2},\ -z\right]$	[6]
7	$\left[\frac{3}{4}-y,\ \frac{1}{4}-x,\ z+\frac{1}{4}\right]$	[7]
8	$\left[y+\frac{1}{4},\ x+\frac{1}{4},\ z+\frac{1}{4}\right]$	[8]
9	$\left[\frac{1}{4}-y,\ x+\frac{3}{4},\ z+\frac{1}{4}\right]$	[9]
10	$\left[y+\frac{3}{4},\ \frac{3}{4}-x,\ z+\frac{1}{4}\right]$	[10]
11	$\left[x+\frac{1}{2},\ \frac{1}{2}-y,\ -z\right]$	[11]
12	$\left[\frac{1}{2}-x,\ y,\ -z\right]$	[12]
13	$\left[y+\frac{3}{4},\ \frac{1}{4}-x,\ \frac{3}{4}-z\right]$	[13]
14	$\left[\frac{1}{4}-y,\ x+\frac{1}{4},\ \frac{3}{4}-z\right]$	[14]
15	$\left[-x,\ y,\ z+\frac{1}{2}\right]$	[15]
16	$\left[x,\ \frac{1}{2}-y,\ z+\frac{1}{2}\right]$	[16]
17	$\left[x+\frac{1}{2},\ y+\frac{1}{2},\ z+\frac{1}{2}\right]$	[17]
18	$\left[\frac{1}{2}-x,\ -y,\ z+\frac{1}{2}\right]$	[18]
19	$\left[y+\frac{1}{4},\ x+\frac{3}{4},\ \frac{3}{4}-z\right]$	[19]
20	$\left[\frac{3}{4}-y,\ \frac{3}{4}-x,\ \frac{3}{4}-z\right]$	[20]
21	$\left[\frac{1}{2}-x,\ \frac{1}{2}-y,\ \frac{1}{2}-z\right]$	[21]
22	$\left[x+\frac{1}{2},\ y,\ \frac{1}{2}-z\right]$	[22]
23	$\left[\frac{1}{4}-y,\ \frac{3}{4}-x,\ z+\frac{3}{4}\right]$	[23]
24	$\left[y+\frac{3}{4},\ x+\frac{3}{4},\ z+\frac{3}{4}\right]$	[24]
25	$\left[\frac{3}{4}-y,\ x+\frac{1}{4},\ z+\frac{3}{4}\right]$	[25]
26	$\left[y+\frac{1}{4},\ \frac{1}{4}-x,\ z+\frac{3}{4}\right]$	[26]
27	$\left[x,\ -y,\ \frac{1}{2}-z\right]$	[27]
28	$\left[-x,\ y+\frac{1}{2},\ \frac{1}{2}-z\right]$	[28]
29	$\left[y+\frac{1}{4},\ \frac{3}{4}-x,\ \frac{1}{4}-z\right]$	[29]

continued ...

Table 7

No.	position	mapping
30	$\left[\frac{3}{4} - y,\, x + \frac{3}{4},\, \frac{1}{4} - z\right]$	[30]
31	$\left[\frac{1}{2} - x,\, y + \frac{1}{2},\, z\right]$	[31]
32	$\left[x + \frac{1}{2},\, -y,\, z\right]$	[32]